Bitcoin

Cryptocurrencies such as Bitcoin are generated online. They can be transferred directly from one person to another across the internet, or bought with and sold for conventional currency.

How it works

Launched in 2009, Bitcoin was the first cryptocurrency and is still the most widely used. Unlike conventional currency, which is backed by the state, Bitcoin is based on cryptography, a system that creates mathematical codes to provide high levels of security. This makes it almost impossible for anyone to spend funds from another user's digital wallet, or to disrupt transactions, as each transaction needs to be verified by other users. This is done by users who transform transactions into pieces of unique digital code. The users who verify transactions in this way are rewarded with bitcoins. Coins can be traded on digital exchanges or by individuals.

A bitcoin transaction

Bitcoin users first set up a virtual "wallet". This acts like a highly secure online bank account for sending, receiving, and storing bitcoins. Wallets can be linked to a conventional bank account to transfer fiat money in exchange for bitcoins. When bitcoins are moved from one wallet to another, individuals in the Bitcoin network, called "miners", compete with one another to be the first to verify the transactions. Once verified, the funds will appear in the recipient's Bitcoin wallet. Buyers usually pay a small fee of about one per cent, which is distributed among the miners.

1. Buyer pays seller for item

A buyer pays a seller in bitcoins using an online transaction form, stating the wallet ID of the seller and the amount to send. The transaction becomes visible to the buyer, and to everyone on Bitcoin's network. It contains a secret code called a private key, showing which wallet it is from. It is grouped with other transactions from a set period into an encoded list called a "block".

2. Mining for bitcoins

To prove that the transactions contained in the block are legitimate, and do not contain coins that have already been spent elsewhere, bitcoin miners use computer programs to solve the complex mathematical puzzles protecting the block. The first miner to do so is rewarded with new bitcoins – which is how new currency is issued – and the block joins the blockchain.